

Prop schwartz imp lp

Proposición 1. *La clase de Schwartz $\mathcal{S}(\mathbb{R})$ es un espacio vectorial sobre $\mathbb{C}$ contenido en $\mathcal{L}^p(\mathbb{R})$ para todo $p \in [1, \infty]$.*

Demostración: Que $\mathcal{S}(\mathbb{R})$ es un espacio vectorial es fácil de comprobar. Sea $f \in \mathcal{S}(\mathbb{R})$, entonces $\rho_{0,0} = \sup_{x \in \mathbb{R}} |f(x)| < \infty$, luego $f \in \mathcal{L}^\infty(\mathbb{R})$. Por otro lado, para $p \in [1, \infty)$,

$$\|f\|_p^p = \int_{\mathbb{R}} |f(x)|^p dx = \int_{|x| < 1} |f(x)|^p dx + \int_{|x| \geq 1} |f(x)|^p dx.$$

Como $f \in \mathcal{L}^\infty(\mathbb{R})$, la primera integral está acotada por $2 \|f\|_\infty^p$. Para la segunda integral,

$$\int_{|x| \geq 1} |f(x)|^p dx = \int_{|x| \geq 1} \frac{|x|^{2p} |f(x)|^p}{|x|^{2p}} dx \leq \rho_{2,0}^p \int_{|x| \geq 1} \frac{1}{|x|^{2p}} dx = \rho_{2,0}^p \cdot 2 \int_1^\infty \frac{1}{x^{2p}} dx < \infty.$$

Por tanto, $\|f\|_p < \infty$ y $f \in \mathcal{L}^p(\mathbb{R})$. ■

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